

# Efficacy of a pain neuroscience educational program in improving executive function and pain intensity in fibromyalgia patients (Fibrobrain project): Study protocol for a randomized controlled clinical trial

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## ABSTRACT

**Background:** Fibromyalgia (FM) is a chronic musculoskeletal condition typically characterized by chronic pain, sleep disturbances, chronic fatigue, and cognitive problems. The present study aims to examine the efficacy of a pain neuroscience education program (PNE) in improving executive functions and pain intensity in patients with FM and compare them with those of standard treatment protocols.

**Methods:** This blinded, controlled clinical trial will compare changes in executive function and pain intensity between two groups. The first group will receive standard treatment supplemented with PNE, consisting of ten sessions spread over five weeks, with each session lasting 40–45 min. The main objective of this method is to remodel cognitive and emotional reactions to pain, challenging the idea that pain is directly related to injury and highlighting the influence of emotions, sleep, and physical activity on pain perception. The second group will receive only standard pharmacological treatment. The study sample will include 62 adults diagnosed with FM, according to initial sample size estimates based on previous evidence. A baseline assessment of baseline characteristics will be performed, after which patients will be randomly assigned to the PNE group or the control group. The results of the intervention will be evaluated and statistically compared after 5 weeks. This protocol complies with all relevant ethical guidelines. It has been approved by two institutional committees (Reference: NR2006; Conbioethics:21-CEI-004-20,170,829).

**Conclusion:** It is anticipated that this intervention will be a cost-effective and superior alternative to standard treatments.

Clinical trial registration: NCT05084300

## 1. Background

Fibromyalgia (FM), a chronic musculoskeletal condition with a global prevalence rate of 2.7 % [1], poses a significant burden on health and social care services in industrialized countries [2,3]. It is typically

characterized by chronic widespread pain, disturbed sleep, fatigue, depression, and cognitive dysfunction affecting the patient's concentration and memory (referred to as “fibrofog” by patients and as “dys-cognition” in the medical literature) [4,5]. Performance-based tests examining dyscognition in FM patients have identified deficits in

**Abbreviations:** (FM), Fibromyalgia; (PNE), pain neuroscience education program.

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working memory, executive control, and attention [6–9], and these are particularly pronounced when performing tasks requiring heightened levels of attention (e.g., in the presence of distractions). However, performance disparities are not consistently evident in tasks with lower demands [10].

Several studies comparing FM patients and healthy controls (HC) using neuroimaging techniques have reported differences in brain activity during painful stimulation, suggesting that the former exhibit heightened activity in the cortical areas associated with pain perception when compared to the latter [11–13].

Pain modulation may be influenced by an increase in facilitation and a decrease in inhibition [14]. Functional magnetic resonance imaging in FM patients shows alterations in brain activity, with notable increases in the insula, cerebellum, amygdala, and thalamus, and correlations between cerebellar activation and catastrophic scores [15–17].

Several cortical regions including the prefrontal cortex, anterior and mid-cingulate cortex (MCC), supplementary motor area, and insular cortex play a significant role in pain processing as well as other sensory and cognitive functions [18]. Previous studies suggest that FM patients may experience cognitive difficulties due to heightened engagement of these overlapping regions of the brain when processing painful stimuli [19]. Furthermore, diminished activity in the brainstem suggests complications with pain regulation in FM patients [20]. Other studies emphasize a thalamic dysfunction associated with chronic pain syndromes [14,21–23], indicating potential alterations in the endogenous opioid system, potentially exacerbating hyperalgesia [24–26].

Glass et al., [10] suggested that FM patients exhibited unique brain activation patterns during executive tasks when compared to HCs, particularly in areas linked to inhibition and attention. These disparities may be attributed to shared neural pathways between pain perception and inhibitory tasks, highlighting the functional plasticity of FM and suggesting potential brain activity normalization and the need for alternate brain regions due to resource constraints. Schmidt-Wilcke et al., [27] observed a correlation between clinical pain and cingulate cortex activation in FM patients performing inhibitory tasks, highlighting the interwoven nature of pain and cognition. FM patients often experience cognitive challenges, potentially due to central sensitization syndrome that amplifies neural signals and pain sensitivity [28–30].

Pain Neuroscience Education (PNE), a contemporary educational approach, aims to modify cognitive and emotional responses to managing a patient's pain, thereby challenging the belief that pain and injury are directly linked and highlighting the role of emotions, cognition, and physical activity in pain perception [31–36]. Pain Neuroscience Education (PNE) is grounded in the biopsychosocial model, which recognizes the intricate interplay between cognitive, emotional, and physical factors in pain perception and management. The integration of essential psychological components is crucial for effective pain management, as highlighted by Sharpe et al. [37]. In this trial, PNE is evaluated as a complementary approach to standard fibromyalgia treatment, which includes pharmacotherapy and physical activity guidelines. PNE is not intended to be a standalone treatment but rather an adjunctive intervention aimed at enhancing the overall effectiveness of the standard management plan for FM. The neuroplasticity of the brain allows for cognitive restructuring when patients are educated about pain mechanisms, potentially leading to improvements in executive functions. Despite the complexity of executive functions, educational interventions like PNE hold promise for influencing these cognitive domains, especially in chronic pain conditions such as fibromyalgia. This trial aims to provide empirical reevidence on whether and how PNE can affect cognitive functions, positioning our study as a crucial step in advancing patient care in fibromyalgia [38]. Evidence suggests that PNE is effective in improving the symptoms and quality of life of patients with chronic pain conditions such as FM [31] [39–41]. However, few studies to date have examined the efficacy of PNE in treating FM [42–45], and none have assessed its impact on the executive functions of patients. Given that PNE is a low-cost intervention that is

beneficial for individuals with chronic pain [31,39,40] and can be easily implemented in various settings, studies exploring this intervention in the context of FM treatment should be conducted.

The novelty of this trial lies in its exploration of the impact of PNE on executive functions in fibromyalgia, an area that has been scarcely investigated. While PNE is known for its pain-reducing effects, its influence on cognitive functions in fibromyalgia remains largely unexplored [46]. This study aims to provide empirical data on how PNE, based on the biopsychosocial model, may enhance cognitive functions through neuroplasticity, representing a significant advancement in the holistic treatment of fibromyalgia.

The present single-blinded randomized controlled clinical trial aims to examine the efficacy of PNE in improving executive functions and pain intensity of FM patients and compare them with that of standard treatment protocols. The hypothesis being tested is that PNE interventions targeting cognitive and emotional responses related to pain will improve executive functions and reduce pain intensity in FM patients more effectively than standard treatment protocols.

## 2. Material and methods

### 2.1. Trial design and context

A randomized controlled parallel-group clinical trial with researcher blinding will be carried out. Recruitment will take place between January 2023 and January 2024 at the Instituto Transdisciplinario en Rehabilitación in Puebla, Mexico and the Centro de Artritis y Osteoporosis in Barranquilla, Colombia.

The proposed trial will be conducted in accordance with the SPIRIT guidelines for clinical trial protocols [47] and the CONSORT guidelines for clinical trials [48] using the CONSORT-EHEALTH [49] and outcomes [50] extensions. The trial has been registered with [ClinicalTrials.gov](https://clinicaltrials.gov) (Identifier: NCT05084300).

This document is Version 1.0 of the study protocol, dated August 7, 2024. Any future amendments will be documented with the corresponding version number and date. All versions will be archived, and a change log will be maintained to keep participants and stakeholders informed of the latest version.

### 2.2. Eligibility criteria

We intend to enroll 62 adult women with FM. The cause and pathogenic process of fibromyalgia remain incompletely understood. Both genetic and environmental factors, as well as peripheral and central mechanisms, are believed to contribute to the widespread pain and heightened pain sensitivity experienced by individuals with FM [51]. As such, diagnosis is based on the clinical criteria described by the American College of Rheumatology and have been revised in recent years [52–54]. In 1990, the American College of Rheumatology created the FM classification criteria to aid in the diagnosis and standardize patients for clinical trials. This criterion is based on the presence of diffuse pain present for at least 3 months associated with 11 tender points (the palpation of the points was removed). However, in 2011, these criteria were revised to encompass other important aspects of fibromyalgia, such as fatigue, unrefreshing sleep, cognitive symptoms, headache, depression, and abdominal pain. In 2016, a revision of the 2010/2011 criteria was proposed to correct classification errors observed in patients with regional pain, a complementary criterion of diffuse pain was added [51,54,55].

The study will include female patients aged 18 years or older who meet the ACR 2010 diagnostic criteria for FM (American College of Rheumatology) [50].

### 2.3. Inclusion criteria

- Those who provide informed consent.

- Those able to communicate effectively with the team.
- Those scoring  $\geq 26$  points on the Mini Mental State Examination.
- Those with at least 12 years of education (elementary and secondary school).
- Those whose native language is Spanish.
- Those with access to the internet and a computer.

## 2.4. Exclusion criteria

- Women who are pregnant.
- Patients with major neurological or psychiatric conditions, intellectual disabilities, premorbid learning disorders, or significant language comprehension issues.
- Patients diagnosed with rheumatic diseases.
- Patients scheduled for surgery within the next 3 months.
- Individuals with recent traumatic injuries.
- Individuals diagnosed with fractures within the past one month.

## 2.5. Participant selection, recruitment, and consent

The study sample will be recruited using the patient records of rheumatologists and only those who meet the eligibility criteria will be included. Following preliminary assessment by a member of the research group and subsequent discussions, the primary care physician reached out to potential candidates for recruitment. All eligible patients were provided with a detailed document describing the study and were requested to complete an informed consent form that had been sanctioned by all participating institutions. Once consent had been collected, a preliminary assessment was undertaken at baseline before randomly allocating the patients to the study groups. A second assessment was performed after 11 weeks. A flow chart of the study has been shown in Fig. 1 and Appendix 1.

### 2.5.1. Participant retention strategies

To minimize dropout and ensure high retention, we will use regular reminders, flexible scheduling and personalized follow-up calls to address concerns and support participant engagement.

## 2.6. Sample size

The sample size was determined based on evidence from previous research [56,57] and using a *t*-test with the Wilcoxon-Mann-Whitney method for two groups on GPower 3.1.9.6. An a priori analysis with a two-tailed design and normal distribution was used to set an effect size

of 0.75, an  $\alpha$  error of 0.05, and a power of 0.80. The findings suggested a sample size of 31 patients per group (total = 62 patients) with an N2/N1 allocation ratio of 1.

## 2.7. Allocation and randomization

Patient allocation will be carried out by an independent researcher not involved in data acquisition or statistical analysis using a 1:1 ratio on the GraphPad 2018 software (GraphPad Software, San Diego, USA) to minimize selection bias. This ensures fully automated random assignment, thereby preventing any influence on group allocation by participants or researchers. Additionally, the research team responsible for statistical analysis will be blinded.

## 2.8. Blinding

To minimize the risk of information bias during patient tracking, the lead researcher will be blinded to group allocations and all evaluations postintervention will either be self-administered under a physiotherapist's guidance or handled by a third-party with expertise in the area. Additionally, patients will also be instructed not to disclose their group affiliation to ensure continued evaluator blinding. Distinct therapists will oversee the evaluation and intervention sessions to ensure impartiality.

## 2.9. Harms monitoring and reporting

An independent Data Monitoring Committee (DMC) will be established to oversee the safety and efficacy of the trial. The DMC will meet quarterly to review interim data, monitor adverse events, and ensure the trial's integrity. The committee will have the authority to recommend modifications or termination of the trial based on safety concerns or data trends. The DMC will provide regular reports to the Ethics Committee and principal investigators.

## 2.10. Intervention group

The intervention group participated in a PNE program based on an educational model designed to teach the biopsychosocial aspects of pain. This model drew from various educational interventions [58] and included information on Explaining Pain [59] [30], Therapeutic Neuroscience Education [60], and PNE [31].

A certified professional (MM) delivered a didactic educational model via live-stream to groups of 4–5 patients using a digital platform, and the

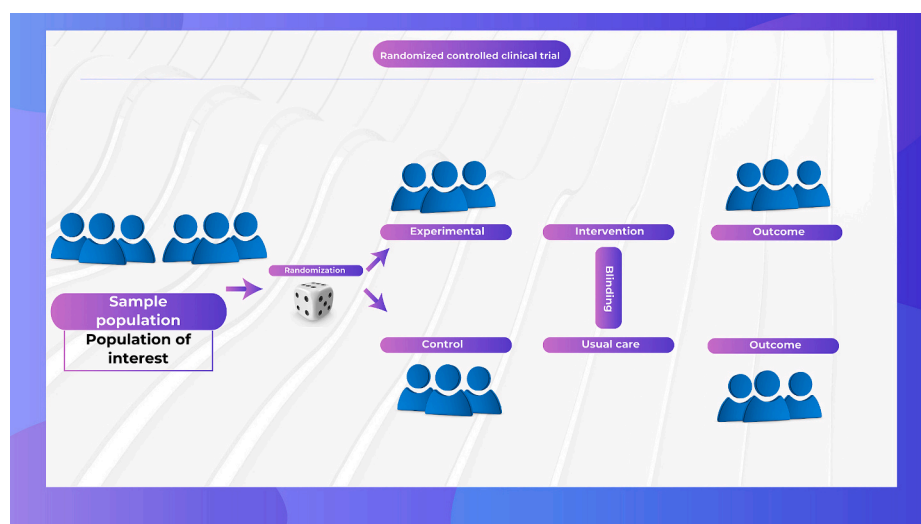


Fig. 1. Flow chart of the study.

sessions were tailored according to previous studies and guidelines [61,62]. After each session, all questions from the patients were addressed, and a comprehensive summary of these sessions have been provided in Table 1.

The sessions were archived on a virtual platform where additional resources were also made available for access (<https://proyectofibrobain.org/>). The content of the sessions was enhanced using illustrations from the second edition of the Spanish version of a book on pain [27]. This content aligns with materials from <http://www.paininmotion.be/> [32], Louw’s manual [33], and explanations of pain in FM [63]. The material was further enhanced using examples and metaphors, and organization of the contents of this intervention was carried out in distinct phases [64]. Group PNE sessions were held twice a week for a duration of 5 weeks, with each session lasting approximately 40–45 min.

2.11. Control intervention

Patients in both the intervention and control groups received standard FM treatment, which includes pharmacotherapy and physical activity guidelines. In addition, specific guidelines for physical activity are reinforced during the PNE sessions, emphasizing the importance of movement and physical activity in fibromyalgia management, as part of a comprehensive multimodal approach. In Latin America, the standard treatment protocols for FM patients primarily involves pharmacological therapy tailored to the individual’s symptomatic profile. This often includes Selective Serotonin Reuptake Inhibitors, other antidepressants,

opioid analgesics, and membrane stabilizing agents. Patients were instructed not to alter their medication regimen during the 8-week period.

2.12. Community and expert participation

The sessions were structured and the workbook developed in collaboration with patients who were board members of an FM foundation in Mexico City. We also sought guidance from FM experts to refine and consolidate the chapters, ensuring we addressed the emotions and concerns that surfaced during the explanation.

2.13. Outcomes

The outcome measures of interest were based on the pain assessment guidelines published in the IMMPACT recommendations [65]. Patient demographic data including age, height, body weight, body mass index, education level, and duration of symptoms were collected, and evaluation of pain intensity, quality of life, cognitive flexibility, cognitive inhibition, working memory, short-term memory, central sensitization, catastrophizing, fear of movement, depressive symptoms, and anxiety was carried out.

The primary outcome measure was the intensity of pain, measured using the Visual Analog Scale-Pain (VAS) and Revised Fibromyalgia Impact Questionnaire (FIQR). Additionally, executive functions were assessed using the Wisconsin card sorting test (WCST), Simon task (ST), Corsi Block-Tapping task (CT), and the Digit Span Task (DST). Assessments were carried out at baseline and 6 weeks after intervention.

2.14. Secondary outcome measures

The secondary outcome measures included other important symptoms of FM, assessed using the Central Sensitization Inventory, Pain Catastrophizing Scale (PSC), Tampa Scale for Kinesiophobia-11 (TSK-11), Beck Depression Inventory, Anxiety Inventory, and State-Trait Anxiety Inventory (STAD).

2.15. Data analysis

A visual representation of the patient’s journey from recruitment to follow-up will be developed in accordance with the CONSORT guidelines. This diagram will highlight the size of the study sample, rationale for any exclusions, the number of patients assigned to various study segments postrandomization, and any discontinuations or disruptions within the intervention groups [47]. Interim analyses will be conducted at specific milestones by an independent Data Monitoring Committee (DMC) to ensure safety and efficacy, with predefined criteria guiding the continuation, modification, or termination of the trial.

A preliminary data analysis including assessment of data discrepancies, biases, anomalies, incomplete or missing data points, normality of distribution, linearity, and homoscedasticity will be carried out to validate essential prerequisites required for further statistical analysis.

The patient’s sociodemographic characteristics and clinical features, particularly those related to the underlying chronic conditions, will be examined at baseline. Descriptive statistics including central tendency measures, variance metrics, frequency charts, and 95 % confidence intervals will be generated for all subgroups. Similarities between groups will be examined using tests appropriate for the type of variable. A univariate analysis of each variable before and after intervention will be carried out. Additionally, bivariate analyses will be used to examine the correlation between dependent and independent variables, while a multivariate analysis will be used to examine the interactions and relationships between them.

The primary outcome measures will include pain scores [evaluated using the Visual Analog Scale-Pain (VAS) and the Revised Fibromyalgia Impact Questionnaire (FIQR)] and executive function scores [measured

Table 1  
Organization of the session protocol.

| Session                                                       | Objective                                                                                                                         | Final Question                                                                           |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1) Introduction to fibromyalgia and the biopsychosocial model | Provide an overview of fibromyalgia and introduce the biopsychosocial model as a reference framework.                             | How do the biological, psychological, and social components interrelate in fibromyalgia? |
| 2) Pain: from sensation to perception                         | Explain the difference between pain sensation and perception, and how the latter is modulated by emotional and cognitive factors. | Why doesn't pain always indicate tissue damage?                                          |
| 3) The brain and the alarm system                             | Address how the brain interprets pain signals and activates the alarm system.                                                     | How do past experiences affect current interpretation of pain?                           |
| 4) Sensitization of the nervous system                        | Introduce the concept of sensitization and how it relates to symptoms like allodynia and hyperalgesia.                            | What is central sensitization and how does it contribute to fibromyalgia?                |
| 5) Stress, emotions, and their impact on pain                 | Discuss the role of stress, emotions, and neuroplasticity in pain modulation.                                                     | How can stress intensify pain perception?                                                |
| 6) The influence of beliefs and thoughts on pain              | Analyze how the patient's beliefs, attitudes, and thoughts can influence their pain experience.                                   | How can their beliefs about pain change?                                                 |
| 7) The role of movement and physical activity                 | Emphasize the importance of movement and physical activity in pain and fibromyalgia management.                                   | Why is it essential to stay active despite the pain?                                     |
| 8) Coping strategies and self-care                            | Offer tools and strategies based on self-awareness and self-care to cope with pain.                                               | Which tools or strategies do you find most useful for managing your pain?                |
| 9) Learning integration and goal planning                     | Review what has been learned and assist patients in setting personal goals.                                                       | What changes do you plan to implement based on what you have learned?                    |
| 10) Future strategies and closure                             | Summarize the program and discuss how patients can continue to apply what they have learned in their daily lives.                 | How do you envision your path to pain management in the future?                          |

Own Source.

using the Wisconsin card sorting test (WCST), Simon Task (ST), Corsi Block-Tapping Task (CT), and Digit Span Task (DST)]. The secondary outcome measures will include the quality of life scores, the Central Sensitization Inventory, Pain Catastrophizing Scale (PSC), Tampa Scale for Kinesiophobia-11 (TSK-11), Beck Depression Inventory, Anxiety Inventory, and State-Trait Anxiety Inventory (STAI). The student's *t*-test will be used to examine differences in mean values between groups, while the paired *t*-test will be used to examine changes within groups. Effect size interpretations will be carried out using the Cohen's *D* index or coefficient, grouping them into small (0.2–0.49), moderate (0.5–0.8), or large (>0.8).

The multivariate analysis will include linear regression for dependent variables such as pain intensity, quality of life, cognitive flexibility, cognitive inhibition, working memory, short-term memory, central sensitization, catastrophizing, fear of movement, depressive symptoms, and anxiety. If the criteria of normality, homoscedasticity, and variable independence are not met, the possibility of transforming the variable will be evaluated. Upon completion of regression, we will assess whether the model represents or resembles the observed data. The models will be adjusted for potential confounders including pharmacological dosage, presence of adjuvant treatment, and previous hospitalization for pain. Additionally, the possibility of a change in the effect of these variables will be assessed.

The evaluation will be carried out in accordance with the established protocol, focusing on those who completed the intervention. However, we will also project our analysis encompassing the entire group with an intention-to-treat approach.

#### 2.16. Data security and management

Participant data will be securely stored in an encrypted database with access restricted to essential research team members. Data will be anonymized before analysis to ensure confidentiality. Regular independent audits will be conducted every six months to verify compliance with protocols, focusing on data integrity and participant safety. The principal investigator will review audit findings, and corrective actions will be implemented as needed.

#### 2.17. Test monitoring

A preliminary study will be carried out to ascertain the appropriate dosages for the various surveys being used and to gauge patient response time and capability.

The resources utilized during the intervention will be evaluated by experts and patient advocates to ensure clarity and comprehension of each outlined section.

#### 2.18. Ethical approval

Participants will receive comprehensive follow-up care after the trial, including access to the study results and personalized recommendations for the ongoing management of their condition. If the intervention proves beneficial, participants in the control group will be offered the same intervention after the trial concludes. Additionally, post-trial monitoring will be implemented to identify and manage any potential long-term effects, ensuring continued participant safety and well-being.

All participants will be thoroughly informed about the study's objectives, methods, potential risks, benefits, and their rights, including the right to withdraw from the study at any time without coercion. Informed consent will be obtained from all participants prior to their involvement in the study.

This project received ethical approval from the Academic Committee of the Doctorate in Applied Cognitive Neuroscience at Universidad de la Costa in Barranquilla, Colombia, on June 16, 2021. Subsequent approval was granted by the Research Ethics Committee of the Centro Mexicano

Universitario de Ciencias y Humanidades on September 21, 2021 (Reference: NR2006; Conbioética: 21-CEI-004-20,170,829).

#### 2.19. Patient and public involvement

Given the nature of the intervention, preparatory efforts were made including validation of the contents of the manual by a subset of patients sharing similar traits. This was done to ensure that the contents of each chapter resonated with the patients and was clear to them. After every session, queries regarding the core ideas proposed in the chapters were discussed. The curated manual will be made accessible for reference, and open sessions will be hosted to enable the public to share and discuss the findings.

### 3. Results

Regarding the dissemination of results the research team is dedicated to comprehensive publication of the study findings in high impact peer-reviewed journals and in accordance with the CONSORT guidelines. Additionally, one component of the research is focused on social engagement and dissemination of knowledge among patient representatives and various chronic pain committees.

### 4. Conclusions

This randomized controlled clinical trial expects to incorporate patients from various parts of Latin America, assessing executive function and pain intensity. It is anticipated that this intervention will be a cost-effective and superior alternative to standard treatments, with the potential to redefine FM management and significantly improve patient well-being.

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#### CRediT authorship contribution statement

**Marco Antonio Morales Osorio:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robinson Ramirez-Velez:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Investigation, Formal analysis, Conceptualization. **Johana Mejía-Mejía:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Laura Aline Martinez-Martinez:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Fabian Román:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Pedro Javier Lopez-Perez:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **Leidy Tatiana Ordoñez-Mora:** Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

#### Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

Leidy Tatiana Ordoñez-Mora reports financial support was provided by Santiago University of Cali. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



## Data availability

No data was used for the research described in the article.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cct.2024.107731>.

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